

Several assumptions were made for this deliverable to optimize the model and improve its accuracy:

- A distinction was made between an entire property (such as a house or apartment) and renting out a single room to avoid data duplication (e.g., the address).
- User accounts can serve both hosting and guest functions.
- Property entries include a general description of what they offer or contain.
- Amenities are managed via a standardized list within the system.
- A `Listing\_photos` table was created to group all images related to a property, including an option to designate a featured photo using a binary variable.
- Functionality was included to allow users to submit reports and moderators to review them.
- Only individuals who have scheduled a visit have the option to leave a review.
- In addition to the property description, the model allows for specifying the number of bathrooms and bedrooms, as well as a binary variable indicating whether shared spaces are included.

Entities not explicitly mentioned in the instructions were also added:

- `Listing\_photos`: Manages a set of photos for a listing, allowing one to be selected as the primary image.
- `Property\_amenities`: Manages the set of features or amenities a property possesses.
- `Visits`: Coordinates property visits by storing key visit details.
- `Saved\_listings`: Allows users to save specific listings.

Application lifecycle:

The application status changes according to its current stage in the lifecycle.

1. Submitted and waiting the host to answer it (Pending).
2. Selected the host for further consideration and a possible visit (Shortlisted).
3. When a visit has been scheduled (Visit arranged).
4. Chosen by the host for the room (Accepted) / Not selected by the host (Reject).
5. Withdrawn by the seeker when is no more interested (Withdrawn).

Listing lifecycle:

1. Creation phase (Draft).
2. Active in public searches (Published).
3. Temporarily hidden (Paused).
4. Unpublished without selection (Closed).
5. Successfully filled and by an accepted applicant (Rented).